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An economic approach for improving requirements negotiation models with inspection

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Abstract Stakeholder goals identified during requirements elicitation are usually informal and incomplete statements about a system considered for development. There are numerous approaches for capturing such informal models. For example, we have found the EasyWinWin requirements negotiation method to be an efficient way for attaining consensus among the success-critical stakeholders. The WinWin negotiation model captures stakeholder goals as win conditions, issues, options and agreements. When such a model has to be transformed into more formal representations, quality becomes particularly important. Approaches for validating such informal models can increase quality and provide guidance for further refinement of requirements. Inspection is a proven approach to identify defects and is also applicable to early life cycle artifacts. This paper reports on an empirical study demonstrating the usefulness of an inspection technique for requirements negotiation models. The study employs a conservative economic model, which considers the effect of defect slippage during development on defect detection benefits from inspection. The main finding of the study is that inspection is an economic validation technique for requirements negotiation models. There are, however, certain limitations that need to be studied in more detail.

Keywords Inspection techniques · Requirements negotiation · Stakeholder-based reasoning · Validation of best-practice models

1 Introduction

Involving success-critical stakeholders is recognized as an important principle to increase the likelihood of project success. For example, the WinWin spiral model of software development [1, 2, 3] requires the involvement of stakeholders in each spiral cycle to identify and reconcile stakeholder objectives. The WinWin negotiation approach uses win conditions, issues, options and agreements [4] to build the negotiation model. However, the outcomes of such interactions among stakeholders are typically informal models of the system to be developed. Especially in project inception these models lack precision and contain defects. Evolving and refining such incomplete models into more formal representations or deriving plans from such models can be risky and challenging. As negotiation models provide the rationale for requirements definition and represent an important input to actual development they should be as clear, correct, consistent and complete as possible with respect to a defined negotiation purpose and project phase to avoid serious negative consequences on contract development, project planning or system architecting [5].

There is a general consensus that low-quality requirements typically have negative impacts on the entire software life cycle: For example, Rule [6] reports on several industrial projects badly affected by poorly understood, ambiguous requirements. Kamsties et al. [7] describe problems related to ambiguities in natural language requirements. The need to focus on quality in requirements engineering (RE) has been pointed out by several authors: El Emam et al. [8] describe a detailed method to measure the success of RE approaches emphasising three major factors: (a) the cost effectiveness of the RE process; (b) quality of RE products; and (c) the quality of service provided by RE staff. Lauesen and Vinter [9] assess the capability of RE techniques to prevent defects in requirements. Macaulay [10] presents a thorough list of requirements for RE techniques including quality assurance aspects.

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We have developed EasyWinWin, a groupware-supported method that is based on the WinWin negotiation model [11], and have successfully used it in 50+ projects to date. In these projects we have gained considerable experience in building negotiation models by capturing and negotiating stakeholder goals. Typical defects we found are vague or missing system capabilities, missing operational stakeholders, imprecise definition or use of the project language, or unresolved stakeholder conflicts. Finding such defects is necessary to identify and mitigate risks resulting from miscommunication or requirements mismatches. In real-world negotiations we are facing additional challenges affecting quality: For example, success-critical stakeholders might only be available for a very limited time. Also, we found that it is not effective to dampen the enthusiasm of stakeholders and the creative flow of ideas during a negotiation with detailed wordsmithing exercises [12]. A major lesson learnt in projects that used EasyWinWin is that an approach for validating negotiation models would be beneficial.

Software inspection represents a quality assurance technique that can be applied to any document produced in the software development process. Inspection approaches include Fagan inspections [13], Gilb inspections [14] or reading techniques for inspection [15]. Rigorous empirical studies have demonstrated the effectiveness and efficiency of these techniques [16, 17]. Although traditionally there has been a strong focus on code inspection complementing test efforts, researchers found that inspecting requirements documents may even increase the benefits of inspection due to earlier defect removal and saved rework [14]. Based on these and our own experiences [18, 19] we have developed an inspection technique, *Negotiation Model Checking* (NMC), targeted at requirements negotiation models [20]. While a lot of research has been done in inspection and requirements negotiation, the effects of combining these approaches have not yet been investigated thoroughly.

This paper reports on an empirical study we have conducted to investigate the performance of inspection for the validation of requirements negotiation models for a range of specific defect types. We evaluate the cost-benefit of NMC [20] for requirements negotiation models. The study employs a conservative economic model, which considers the effect of defect slippage on benefits. Finding defects early reduces both the size and probability of rework effort from undetected defects. We found using a requirements negotiation approach in combination with inspection can help to lower important risks in certain project types. The remainder of this paper is organised as follows: Section 2 discusses the context and related work for the validation of requirements negotiation models. Section 3 presents a study we carried out to empirically evaluate the cost and benefits of the NMC approach. Section 4 presents the results of the study and Section 5 discusses the results with application in project practice. Conclusions and an outlook on further work round out the paper.

2 Validation of negotiation models

Eliciting goals from stakeholders is an important but non-trivial task. Goals are needed to provide precise criteria for the completeness of a requirements specification; they help to identify stakeholder conflicts; they capture the rationale for requirements, and drive the identification of requirements that support/implement them [21]. The WinWin negotiation model captures stakeholder goals as win conditions, issues, options and agreements. Agreements describe goals where consensus has been achieved among the stakeholders. The validation of a requirements negotiation model is important to improve its completeness, correctness and consistency.

This section discusses the life cycle context for requirements negotiations with a focus on EasyWinWin (EWW); briefly summarises the NMC inspection technique we have developed for EWW and develops evaluation criteria for inspection (e.g., which defect types are found how effectively and whether inspection is worthwhile in a certain context.) We chose EasyWinWin for our study since it has been used in a large number of real-world projects, reliable tool support is available for negotiations, and the output of an EWW negotiation is a list of mutually agreed stakeholders' commitments.

2.1 Life cycle context for requirements negotiations

The context of this work is iterative life cycle models such as the spiral model of software development [1]. The spiral model is a high-level model describing a set of principles that help to manage different kinds of software development projects. Spiral development has been described as a "family of software development processes characterised by repeatedly iterating a set of elemental development processes and managing risk so it is actively being reduced" [22]. The key features of the approach are as follows [22]:

- Key artifacts such as requirements, plans, designs and code evolve concurrently. Consequently one has to deal with low-precision and incomplete models early on in a project where details are added iteratively.
- Life cycle anchor points define the required levels of detail of artifacts for intermediate commitment points. Anchor points therefore also determine quality criteria in different project phases [2].
- Risks determine the level of effort on activities and the required level of detail of artifacts. Because risks influence the direction of a project they need to be identified as early as possible.
- Emphasis on system activities and artifacts to avoid premature suboptimisation on hardware, software or implementation considerations.
- Involvement of success-critical stakeholders to understand objectives, constraints and alternatives in each cycle.

The WinWin Spiral model especially emphasises the principle of stakeholder involvement and complements the original Spiral Model with activities dealing with stakeholder involvement and negotiation. In each spiral cycle the success-critical stakeholders are identified and their goals are elicited and reconciled in a negotiation process. Defects in the resulting negotiation model can result in severe problems; for example:

- missing or ill-defined capabilities leading to unusable systems, unreliable estimates, or infeasible architectures;
- ill-defined interfaces to other systems or to the user, making integration into the target environment infeasible;
- miscommunication caused by language problems due to missing, unclear, or wrong terms;
- hidden conflicts among stakeholders leading to mistrust;
- misunderstood project constraints such as staffing, required technology, budget, schedules causing project delays, frustration and confusion;
- inconsistent or infeasible quality objectives negatively impacting the choice of feasible architecture and unnecessarily increasing development costs.

Although some of these defects might be detected in subsequent anchor point reviews, we believe that a cost-conscious approach for improving stakeholder negotiation models is essential before using the negotiation model to develop project plans, requirements, architectures, etc. Our approach is based on (a) optimising the negotiation process by providing a groupware-supported methodology for negotiation (see Section 2.2) and (b) improving the quality of negotiation results by applying an inspection technique to negotiation results (see Sections 2.3 and 3).

2.2 EasyWinWin negotiation models

We have found EWW to be an effective and efficient method to elicit and negotiate the objectives for a system under consideration. The approach has a strong focus on establishing a shared vision and a set of shared commitments (agreements) among the success-critical stakeholders. EWW helps them to attain consensus by jointly discovering, elaborating, prioritising and negotiating their objectives [11]. The approach adopts group facilitation techniques that are supported by collaborative tools (e.g., electronic brainstorming, categorising, polling). The input to an EWW workshop is typically a mission statement outlining the high-level objectives of a project and a negotiation purpose statement specifying the objectives of a negotiation within a project. An EWW negotiation is typically carried out in a series of workshops (co-located or dispersed), involving all identified success-critical stakeholders. A facilitator follows detailed

guidelines as described in the EWW process guidebook and moderates the negotiation process. The use of a technographer in charge of handling the groupware tools has proven to increase efficiency especially for large groups.

EWW builds on the WinWin negotiation model [4] and captures stakeholder goals as *win conditions*, *issues*, *options*, *agreements* and *terms*. Major elements of an EWW negotiation model are (1) negotiation topics organised in a domain taxonomy, (2) definitions of key project terms, (3) agreements providing the foundation for further plans, (4) open issues addressing constraints, conflicts and known problems, as well as (5) further decision rationale showing the negotiation history (like associated comments, win conditions, issues and options). According to our experience typical negotiations models created by 10+ stakeholders result in 300+ brainstorming ideas, 100+ win conditions, 50+ issues, 50+ options and 100+ agreements. Due to this size and complexity systematic quality assurance techniques are needed. Typical faults we experienced are vague or ambiguous statements, missing information, inappropriate level of detail or inconsistent results.

2.3 Negotiation model checking: an inspection technique for EasyWinWin models

After an EWW negotiation we aim at removing defects and determining quality to guide subsequent activities. Inspection is a well-explored technology for the validation of requirements documents [15] and can be adapted to validating negotiation models. The inspection process used in this study follows the standard process and consists of the following stages [17]:

1. *Inspection preparation* to check the entry criteria of completeness and sufficient quality for understanding the inspection context.
2. *Individual reading* supported with reading techniques tailored to negotiation models.
3. *Meeting of inspectors* or some other form of defect collection.
4. *Report and rework* to clarify the issues raised with the author, if possible, or to document the problem and the resolution in a traceable way. Possible rework strategies depend on defect severity. Minor defects can typically be fixed without renegotiation. Major defects often lead to renegotiation and can make settled agreements obsolete.

In the study we focus on the “individual reading” step. The Negotiation Model Checking (NMC) technique adopts a reading technique [20]. Reading is a key activity in defect detection to understand a given software artifact and compare it to a set of expectations regarding structure, content and desired qualities. Ad hoc inspections depend completely on the capability of

the inspector and are not repeatable. Defect detection should thus be supported with reading techniques [15] guiding inspectors through the document and giving instructions on how to perform quality checking. An example of a general reading technique is a checklist listing possible defect types or symptoms to look for in a particular document type, usually independent from a specific notation [23].

Each inspector is asked to perform the defect detection procedure individually and independently. The technique aims at preparing the inspector with an overview and then continues with detailed checks for clarity, correctness, completeness and consistency. The strategy is to support the inspector with easily digestible steps to avoid cognitive overload. The experience of a small-scale feasibility study [20] has encouraged us to improve the NMC technique and to conduct the more rigorous study presented in this paper.

2.4 Evaluation criteria for inspection

Inspection is a quality assurance technique for assessing the quality of a system model. Project managers, for example, can use inspection results for guidance in further development. Inspection activities are an investment that (a) evaluate product quality, development work and the effectiveness of development processes, and (b) save money by eliminating minor and major defects that would cause rework [24]. Major evaluation criteria for the inspection of negotiation models are the effectiveness to find defects in important defect categories, e.g., severe defects rather than minor defects. In addition, cost–benefit analysis helps to determine in what contexts inspection is likely to be worthwhile. Such an analysis balances the invested effort with likely saved staff hours from early defect detection.

2.4.1 Effectiveness criterion

The effectiveness of an inspection process is defined as the ratio of defects found to the total number of defects in the product, or in a given defect type class. In this study we consider three dimensions of defect types: (a) the model part in which the defect occurs (function, interface, quality, glossary or project management); (b) the kind of defect (missing, incorrect, unclear or inconsistent information); and (c) the severity of the defect.

For defect severity we consider the likely impact of a defect on further development: (a) low-severity defects (L) do not considerably increase development effort (compared to a nominal development progress); (b) major defects (M) potentially incur a considerable amount of rework and may increase project risk; and (c) critical defects (C) will most likely cause a lot of rework and/or put the overall project success at risk.

In practice, defect severity often depends on the project stage in which a defect is found and removed.

For example, a missing capability defect that may be easy to fix during requirements definition may become a major problem during implementation if the capability cannot be realised with the chosen system architecture. Thus we distinguish for each defect three cases based on risk expectations for development: the best case (B), a nominal case (N) and a worst case (W), with more or less increasing defect severity depending on the nature of the defect. In practice, the quality manager can track defects in a set of comparable projects to fine-tune the rating of the likely impact of a defect [25].

Inspection efficiency is defined as the number of defects found per person-hour as an initial indicator for process economy.

2.4.2 Net gain criterion

The net gain, a more sophisticated economic indicator than efficiency, is the difference between inspection benefits and inspection costs. An activity not yielding a net gain is not advisable from an economic point of view. In this cost model we use the staff hours invested by a nominal inspection team for reading the inspection object as direct inspection costs. In a real project context further indirect costs would accrue that depend on the local cost structure, e.g., for inspection planning and for the identification of false positives.

The economic benefit of an inspection is the saved future effort for development due to higher quality of inputs for development and project planning. Benefits come from estimated savings of rework if a defect has to be detected and removed later in development or operation. The project manager or developer must estimate the benefit of a defect. This benefit from savings depends on the severity of the defect and the impact it would have had on the development project, which may vary with the development phase in which it would have surfaced.

As indicated above, defects may slip into later development stages and thereby increase the risk/cost to the project. The cost associated with defects of different severity and cases can be expressed in a 3 by 3 table (see the benefit distribution in Table 1 in Section 2.4.2). In this work we distinguish three severity levels of defects and three phases depending on the additional effort to find and fix a defect in a given class, if it is not found during inspection. The benefit of finding a given defect

Table 1 Left: number of seeded defects (in document A/B) by defect severity; right: expected benefit per defect found during inspection

	Defect distribution			Benefit distribution		
	L	M	C	L	M	C
Best case	24/17	16/19	0/5	0.5 h	1 h	2 h
Normal	16/16	19/16	5/9	1 h	4 h	16 h
Worst	14/16	10/10	16/15	2 h	16 h	128 h

depends on the difficulties this defect would have caused in a hypothetical project context. This is not a fixed value but can be modeled with a probability distribution of expected savings. There are several approaches to determine the benefit for a defect of a given severity class:

- a. The simplest approach is to assign a single benefit value to each defect class.
- b. Another approach would be to assume a probability distribution of benefits for each class. This can, for example, be a triangle distribution for best, most likely and worst cases. The expected benefit for a given defect is determined from this benefit distribution [24].
- c. A more sophisticated approach includes benefit estimates for each development phase in which the defect could be detected. In an early phase the impact of a defect is rather low (e.g., in-house design), while in a later phase the impact is much higher, since a defect potentially spreads in related software products and various people are involved in removing the defect (e.g., operation at the customer site).

In this paper we use the third approach. For the assumption on inspection benefits we use rather conservative (low) benefit values to stay on the cautious side in the economic evaluation (see Table 1 for details). The total inspection gain is then calculated as the difference between total inspection benefits (i.e., summing up inspection benefits for all detected defects) and inspection costs (i.e., total effort invested in inspection).

2.4.3 Research issues

The overall research goal is to investigate whether inspection can be an economic approach for the validation of requirements negotiation models. In the context of the evaluation criteria in the preceding subsections we state our expectations for defect types to be found with an inspection in an initial EWW requirements negotiation:

1. We expect the inspectors to find (1a) fewer defects of type “missing” than overall defects, since missing information is harder to imagine than defects in given information, and (1b) an over-proportional number of defects of type “wrong” or “incomplete” as these are rather easy to identify in a given model.
2. We call inspection to be an economical approach for a set of assumptions, if the conservative economic model yields a positive average net gain in a certain context. In our study we look at the contexts (a) individual inspectors and (b) teams of three inspectors, as this team size is reasonable in practice. We expect three-inspector teams to have a higher effectiveness than individual inspectors, but with lower efficiency and hence lower net gain per inspector.

3 Empirical study description

The goal of this empirical study was to evaluate the performance of software inspection for negotiation models. In this section we describe the empirical study, which we performed to evaluate the research questions. We will outline the experimental process, describe the inspected artifacts, the seeded defects, and the participating subjects. Finally, we will discuss threats to validity and present the techniques applied during data analysis.

3.1 Study background and process

The empirical study was conducted at the Vienna University of Technology as part of a workshop on quality assurance. Thirty-two students participated in the workshop and applied the following experimental process under the supervision of the research team consisting of four members (faculty and graduate students).

In a first step students received background information and theoretical knowledge through lectures and were then trained on requirements negotiation and inspection techniques. At this stage we collected information on software engineering and quality assurance experience for each participant using a questionnaire. In the second step students were randomly assigned in teams of eight and performed a requirements negotiation following the EWW process outlined in Section 2.2. We chose a project for which we could safely assume that students had sufficient domain knowledge: the task was to define the basic functions, interfaces and quality requirements of a Web portal for computer science students at Vienna University of Technology. The system should provide services for students such as curriculum information, course registration, download of teaching material, or online communities. In order to stimulate discussion during negotiation we assigned different roles to the students (e.g., end-users of the Web portal, faculty, administrators).

After completing the negotiation, the research team collected the negotiation results of the four teams, selected the two most complete ones, further improved quality by searching for defects, removed detected defects, and then seeded back a number of reference defects. Two negotiation result documents were generated with a known number of reference defects, which represented the inspection objects A and B. Finally, the inspection was performed in a classroom set-up where all students synchronously inspected one of the two inspection objects individually and independently. The research team ensured that each inspector received an inspection object he was not aware of, i.e., where he had not participated in the negotiation process. Furthermore inspectors received the NMC checklist (see Section 2.3) as guidance for individual defect detection, a defect report form to collect defect information and a feedback

questionnaire in order to provide feedback on the inspection process.

After individual inspection the research team collected all documents and performed defect matching for the paper-based information and created electronic representations of the data for further data analysis.

3.2 Documents

The study participants generated the two inspection objects used during the study (documents A and B in the remainder of this text)¹. Both documents use natural language (German) to represent the negotiation model and are structured as follows: The “*Negotiation Context*” section describes the general purpose and context of the negotiation (this section is identical for documents A and B); the second section contains the *glossary* defining the project language (26 items in both documents) that lists all important terms, which (a) have a special meaning in the negotiation context or (b) require clarification in general (e.g., technical vocabulary); the third section represents all win conditions (42/60 for document A/B), issues and options; the last section summarises all agreements (42/60 for document A/B) and unresolved issues where no agreements could be found (1 for both documents).

3.3 Defects

In order to appropriately evaluate the performance of inspection, we improved the inspection objects and then seeded reference defects back into the document (40/41 defects for document A/B) in a first step. Note that all reference defects were found in the original requirements negotiation model.

As defects at this early development stage can have various impacts on further project progress, we applied a multi-dimensional defect characterisation schema to the defects: The first set of defect attributes describes the area of the system, where the defect would cause damage: function, interface, quality, glossary and project management. The second dimension of the defect classification summarises defect type information indicating descriptions of the defects: unclear, wrong, missing and inconsistent information. Table 2 summarises the distribution of defects for these two dimensions. Note that in this defect classification schema a reference defect can belong to more than one defect class.

The third defect classification dimension covers defect severity (= low, major or critical). Defect severity measures the potential impact of a defect on project progress measured in hours of rework necessary to remove the defect if detected later during development. The amount of saved rework due to early detection represents the

Table 2 Number of seeded defects (document A/B) by defect type

	Unclear	Wrong	Missing	Inconsistent	Sum
Function	6/3	7/9	4/3	7/9	24/24
Interface	4/3	9/5	5/4	12/8	30/20
Quality	3/1	6/10	2/1	3/9	14/21
Glossary	2/1	4/2	0/3	1/1	7/7
Project Management	0/0	0/4	0/1	0/5	0/10
Sum	15/8	26/30	11/12	23/31	

most significant inspection benefits. In the case of negotiation results the assessment of defect severity is especially difficult, as it is hard to anticipate if a defect would be detected during requirements definition or in a later development process. Therefore we distinguish three different scenarios: a best, normal and worst-case scenario. In the best-case scenario all negotiation model defects are detected in the subsequent process of requirements definition, resulting in a low average amount of rework required for removal. In the normal case some of the defects are found early but some are found during later development stages, increasing the average amount of rework necessary. Finally, in the worst case defects are detected very late, resulting in considerable or in the case of critical defects even project-challenging rework effort.

Table 1 summarises the distribution of defects among different defect severity classes and the values assumed for rework effort used in order to calculate inspection benefit. The rework effort values described in Table 1 are rather conservative than overoptimistic. In practice a project manager can replace those values by experience-based and project-related value distributions and perform simulation analysis in order to estimate inspection benefits.

3.4 Subjects

The subjects participating in this study were 32 computer science graduate students (18 inspecting document A and 14 inspecting document B) taking a course on quality assurance. Inspectors had a varying degree of experience and expertise, which was gathered using a questionnaire. About 20% of the participating subjects had professional experience with software development, requirements documents and defect detection in requirements documents. The remainder had heard lectures and done workshops on these topics at university before participating in this study.

3.5 Validity of study

The goal of this study is to document the performance of inspection for informal negotiation model documents. As all empirical studies, this empirical study is subject to

¹These documents are available from the authors on request.

different threats to validity. However, the study follows a very simple study design (i.e., all inspectors apply the same, general inspection technique), making it rather robust to threats like process conformance or selection (i.e., assignment of subjects to treatments). Furthermore, the moderate number of subjects enabled the experiment team to carefully supervise all activities. Remaining threats to validity stem from the subjects, the inspection object and the seeded defects.

3.5.1 Students as subjects

The subjects were students participating in a university course. As pointed out in Curtis [26], students may not be representative of real developers. However, Höst et al. [27] observe no significant differences between students and professionals for small tasks of judgment. According to Tichy [28], using students as subjects is acceptable if students are appropriately trained and the data is used to establish a trend. These requirements are both fulfilled in our case. Nevertheless, we suggest further work to clarify the performance of our inspection technique with professionals.

3.5.2 Inspection documents

The inspection documents were not created during a real requirements negotiation but stem from student negotiations on a hypothetical project. We decided to let participants create the documents, which were then used during inspection, as this guaranteed that inspectors were familiar with the project and the structure of the inspection object. Furthermore, this approach enabled us to use two different documents during inspection to increase the external validity of results. Selecting documents from external negotiations would have seriously threatened the internal validity of results because of the different context of negotiation and inspection participants. Boehm et al. [3] used a similar design for their experiments on requirements negotiation. Therefore we believe that students can sufficiently act for different stakeholders when supervised carefully.

3.5.3 Seeded defects

Finally, the seeded defects represent a potential threat to the validity of results. However, as we used two different documents and seeded different defects in those documents, the confounding effects due to the seeded defects should be within reasonable ranges. Moreover, we have considerable experience with requirements negotiation documents and common defects from negotiations in different environments and found the seeded defects to be very typical in the project context. Finally, different members of the research team performed the seeding of defects for documents A and B and to avoid systematic trends in defect seeding.

3.5.4 False positives

Another potential threat to validity, which is closely related to the seeding of defects, deals with false positives. For our research goals it was necessary to work with seeded defects, which enable us to really assess the impact of any quality assurance technique. However, we emphasise that false positives play a potentially more important role in the case of informal requirements compared to inspections of design or code documents. In fact, due to the incomplete and informal nature of informal requirements documents, many false positives could potentially be true defects. Nevertheless, we defined defects in a narrower sense and focused on evaluating the inspection performance with respect to our seeded defects.

4 Empirical study results

In this section we summarise most important empirical results using mainly descriptive statistics, i.e., mean, standard deviation, confidence intervals and box plots. The goal is to document the inspection performance for individual inspectors and sampled inspection teams of size three (the selection of size three is arbitrary but represents a realistic team size in practice). The team results presented in this section are based on a simple sampling approach (100 iterations), which builds random teams of size three out of the total sample of inspectors (per document). The only restriction imposed on the sampling approach is that all team members must be separate inspectors. Note that this very simple sampling approach suffers from some constraints (e.g., not every sampled team is unique; there is a certain positive probability that two sampled teams are equal, which imposes a certain bias on the distribution). However, as we are not interested in the true distribution of team performance, but in estimators for its means and standard deviation, we decided to use this simple approach.

As far as inspection performance is concerned, we will look at defect detection effectiveness, efficiency and inspection gain. We will compare the results for documents A and B and analyse whether inspector characteristics have any influence on individual performance.

4.1 Defect detection effectiveness and efficiency

The first evaluation criterion is defect detection effectiveness, i.e., the ratio between the number of defects found and the number of defects originally in the document. In this context, false positives (i.e., defect reports not matching reference defects) should be briefly discussed. As expected, the average number of false positives is high in the context of negotiation model inspection: on average, 50% (58%) of defect reports represented false positives for document A (B). A more detailed analysis of these false positives reveals that for

both documents most of those reported issues arise from potentially missing glossary entries. Due to the preliminary nature of negotiation results, a large number of false positives had some constructive contribution for further improvement even though they did not constitute a defect.

Table 3 summarises defect detection effectiveness for individuals and sampled inspection teams of size three for different defect categories (first column). Furthermore, the column called “Seeded” displays the total number of defects in the document for each category and the column “Performance increase” contains the difference between individual mean effectiveness and team mean effectiveness.

Table 3 shows that for document A (B) individuals find on average 41% (32%) of all defects. This indicates satisfying inspection performance even for individual inspectors. For inspection teams the defect detection effectiveness is further increased to 67% (64%) for document A (B). It is interesting to note that individuals performed better for document A, while the performance increase due to team building is larger for document B.

As far as performance standard deviation is concerned, document B suffers from larger standard deviation than document A, indicating that individual and team performance fluctuate more for document B. Another interesting observation deals with inspection performance regarding different defect types. As expected, inspectors found less fewer defects of type “missing” and “unclear” compared to defects of type “wrong” and “inconsistent”. As far as the defect classification by model part is concerned, defects regarding functionality seem to be hard to detect, because in their case defect detection effectiveness is lowest. In contrast, effectiveness is highest for defects in the glossary for document A but one has to be aware of the fact that comparatively few defects were seeded in the glossary and that the standard deviation of effectiveness is also the largest (also for document B). In the case of document B most defects were found in quality requirements. Figure 1 visualises these observations with box plots of the effectiveness distribution for individual inspectors by these defect categories.

Table 4, finally, summarises individual and team defect detection effectiveness for different defect severities and different scenarios. It shows that both for individuals and teams there are only insignificant differences between effectiveness for defects of different severity levels; i.e., critical defects have a similar probability to be detected as low or major defects.

In addition to analysing defect detection effectiveness we evaluate defect detection efficiency, i.e., the number of defects detected per hour. Efficiency is an initial measure to assess inspection economy. For individuals, the average (standard deviation) efficiency for Document A equals 9.6 (3.5) and for Document B 7.4 (3.2) defects per hour. For inspection teams, the mean (standard deviation) efficiency for teams of size three equals 5.0 (0.6) for document A and 4.9 (0.9) for

Table 3 Effectiveness by document and defect types

	Individual		Team									
	Document A					Document B					Document A	
	Seeded	Mean Effectiveness	SD Effectiveness	Seeded	Mean Effectiveness	SD Effectiveness	Seeded	Mean Effectiveness	SD Effectiveness	Performance increase	Mean Effectiveness	SD Effectiveness
All	40	41.1%	12.5%	41	32.1%	14.9%	26.3	67.4	7.6	31.7	63.8	12.6
Function	16	32.6%	12.2%	17	25.6%	16.1%	28.1	60.7	11.9	29.5	55.1	14.9
Interface	21	40.2%	14.3%	13	31.9%	18.6%	27.0	67.2	9.0	34.5	66.4	15.7
Quality	9	39.5%	15.8%	15	44.3%	17.3%	24.8	64.3	12.1	34.4	78.7	12.3
Glossary	6	60.2%	27.5%	4	32.1%	35.9%	25.2	85.4	11.8	38.2	70.3	33.9
Project management	–	–	–	6	16.7%	19.6%	–	–	–	27.0	43.7	23.4
Unclear	11	36.4%	17.6%	5	25.7%	18.3%	27.7	64.1	15.0	34.3	60.0	17.8
Wrong	21	47.1%	13.3%	23	32.9%	16.2%	25.4	72.5	7.2	39.5	65.2	15.4
Inconsistent	16	42.7%	13.1%	26	32.2%	17.3%	24.0	66.7	8.9	33.0	65.2	14.0
Missing	8	27.1%	15.6%	9	25.4%	19.2%	23.9	51.0	16.6	32.2	57.6	20.4

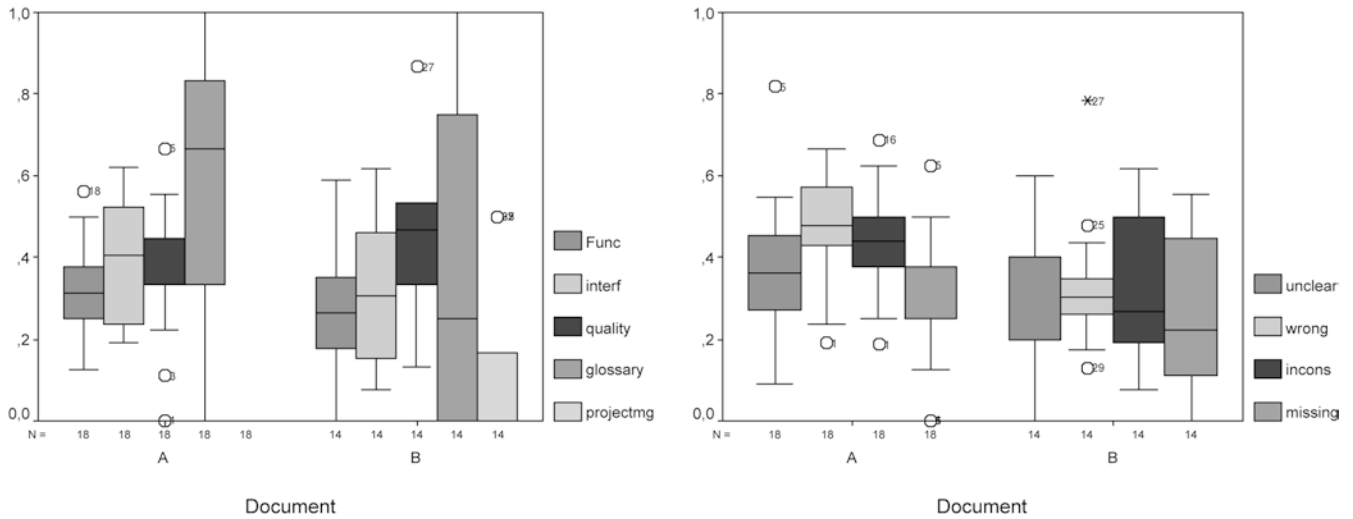


Fig. 1 Box plots of individual effectiveness by defect type

Table 4 Effectiveness (in %) by defect severity and impact case assumption

		Individual						Team					
		Low		Major		Critical		Low		Major		Critical	
		Mean	SD	Mean	SD	Mean	SD	Mean	SD	Mean	SD	Mean	SD
B	Doc. A	38.4	13.6	45.1	14.6	—	—	64.9	8.7	70.4	9.7	—	—
N		36.8	15.9	44.1	13.3	50.0	24.3	61.0	10.7	74.7	8.3	60.4	12.9
W		40.5	17.3	49.4	17.3	46.5	14.6	65.4	10.5	67.0	9.8	69.1	8.3
B	Doc. B	27.3	15.1	34.5	18.3	38.1	28.1	60.4	13.2	69.7	14.0	62.7	29.6
N		26.8	15.6	37.5	17.2	31.7	25.0	59.3	13.8	69.2	14.2	67.0	23.3
W		26.8	15.6	42.9	21.3	30.5	18.2	59.3	13.8	75.3	11.8	61.3	21.0

document B. This is encouraging as these levels of efficiency significantly outperform values reported for other inspection applications [18]. Efficiency decreases for inspection teams but still the observed efficiency levels for teams of size three are remarkable.

Finally, we evaluated the influence of different inspector factors collected using a questionnaire on individual inspection performance. For this data analysis we performed a simple linear regression trying to explain individual defect detection effectiveness by inspection effort and a set of inspector experience variables. However, the analysis did not reveal any significant relationship for document A. For document B, we observed two results: (a) inspection effort has a significant (coeff. 0.42; at the 90% level) positive influence on performance; (b) experience with requirements documents has a slightly positive but significant influence (coeff. 0.19; at the 95% level). However, as the observations are not homogeneous for documents A and B, their general validity is limited.

4.2 Inspection net gain

The last step of data analysis aims at answering the question whether inspection is a cost-effective quality

assurance technique. Therefore we derive inspection gain (i.e., expected inspection benefits minus inspection costs; see also Section 2.4.2 and Table 1 for details) for different defect severity assumptions.

Table 5 summarises the net gain results for individual inspectors and inspection teams. The most important observation regarding this economic analysis is that the net gain is never negative; i.e., even for very optimistic assumptions (i.e., that most defects can also be easily fixed in immediately following phases with little effort) inspection pays off. Inspection teams increase overall inspection net gain by a factor of 1.5 (2) for document A (B) compared to individual inspectors. For some reason, team combination is more beneficial for document B; it seems that in the case of document A there is a higher defect overlap among inspectors (while for document B team combination increases effectiveness).

The most important reason for this interesting result is that inspection of stakeholder goals is extremely efficient compared to inspections of requirements, design or code documents [16, 17, 29]. Therefore inspection costs are comparatively low compared to even conservatively estimated benefits. Note that Table 5 only considers the effort invested by individual inspectors in individual reading as inspection costs. In practice, the project

Table 5 Net gain (in staff hours) for individual inspectors and 3-inspector teams

	Individual								Three-inspector team							
	Document A				Document B				Document A				Document B			
	Mean	SD	90% confidence		Mean	SD	90% confidence		Mean	SD	90% confidence		Mean	SD	90% confidence	
B	10.0	3.5	2.7	15.3	10.4	6.7	1.9	22.6	13.9	2.0	10.5	17.16	18.6	5.7	7.7	26.8
N	75.6	20.0	43.17	111.3	67.9	40.5	21.3	148.2	117.9	15.5	95.3	144.6	126.2	35.0	51.5	175.6
W	760.6	210.5	416.2	1036.5	540.5	362.6	148.4	1314.2	1244.3	136.2	1035.4	1586.1	1068.7	355.2	380	1604.6

manager would have to add some costs associated to inspection preparation and operation. However, these administrative costs should be low compared to the inspection effort.

5 Discussion and conclusion

Requirements negotiation models are informal and incomplete statements about stakeholder goals, constraints and agreements. Such results can be rather hard to formalise and refine. Quality assessment approaches for validating such models can increase model quality and provide data for further refinement in a cost-effective way by using stakeholder time as little as possible. Inspection is a proven quality assurance approach to find defects also in early-life-cycle documents such as requirements specifications. While inspection and requirements negotiation both are well-established research areas, the effects of the combination of these approaches have only been investigated in a first feasibility study [20].

This study focused on the effectiveness of Negotiation Model Checking (NMC) to detect defects on individual and team level and on the cost–benefit performance evaluation. Overall effectiveness was in the range of 30–40% of all defects for most inspectors. This is comparable to reports from inspection experiments with effective reading techniques for requirements specifications [29]. Regarding defect classification by model parts, defects regarding functionality were less often detected than defects in other categories. Project management defects were only present in document B and found with low probability. This finding is consistent with the limited project management background of the study participants. As expected, defects that can be identified without context knowledge (based on the information presented in the document) wrong or inconsistent information were found with above average probability, while defects about missing information were less likely to be found. This is again consistent with the background of the participants, who had taken part in an EasyWinWin negotiation in the domain under study, but lack practical project experience in this particular domain.

The results regarding defect severity are especially interesting. In document A (B) inspectors found 40–50%

(30–40%) of defects of different severity. On average, the technique was for both documents as effective to find major and critical defects as overall defects. The current technique helps to uncover defects of all severity levels with similar probability.

For sampled three-inspector teams the effectiveness for most defect types is in the range of 60–70%, which is encouraging for adopting the approach in practice. In this study the team effect increased performance compared to individual inspectors and diminished differences between documents and defect types. The performance is similarly effective for defects of all severity levels and under all assumptions for defect slippage during development supporting the robustness of the approach. While the effectiveness of the teams is higher than for individuals the efficiency and the performance per inspector both decrease, which is an argument against overly large teams.

Net gain (see Section 2.4.2 for definition) is positive even with conservative defect benefit assumptions for over 90% of all inspectors and teams under all assumptions for defect slippage. An important aspect here is that even the 90% confidence intervals (see Table 5) fully lie within positive ranges, indicating that in the experimental context project managers could be 90% certain of performing an inspection with positive net gain. This very positive result shows that the investigated inspection approach is highly efficient and is very likely to yield substantial savings from rework with the cost–benefit model in this study.

From an industry perspective the main finding of this study is that inspection with the Negotiation Model Checking technique is an economic validation approach. Major limitations of the approach in practice are as follows:

- *Missing information is hard to detect with limited context information.* One limitation of inspection techniques in the context of informal negotiation models is the lack of reference documents or existing knowledge, which can be used to check the inspection object. One possible way to deal with this issue is to focus on group defect detection instead of individual defect detection. Group interactions could potentially prove to be valuable for the detection of certain defect classes, e.g., ambiguities, which are important in natural language requirements.

- *The influence of document characteristics is not entirely clear.* Experiences with real-world EWW negotiations show that such documents are two to three times larger than the documents used in our study. As the study revealed high efficiency for inspection, we expect that an inspection of such larger documents can be conducted in about 1 person-day even with the assumption of exponential growth of document and inspection task complexity. For substantially larger documents further studies would be necessary.
- *The influence of a different defect mix needs to be determined.* In a real-world project we would expect a much higher proportion of defects of the types glossary and especially project management. On the other hand, in a field study we would expect inspectors with a project management background to balance this increase with more expertise in finding such defects than our study participants.
- *Factors for successful inspector selection could not be consistently identified.* While we would in general expect inspectors with project experience to find defects more easily that need context information, we do not yet know which inspector characteristics make a good defect finder.

These issues need to be studied in more detail and the technique should be used in practice with appropriate consideration. Nevertheless, finding defects – especially major and critical defects – early in the software development process is an important success factor and reduces both the size and probability of rework effort stemming from undetected defects.

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